

AI Workspace Platform — Architecture Diagram

System Architecture Overview

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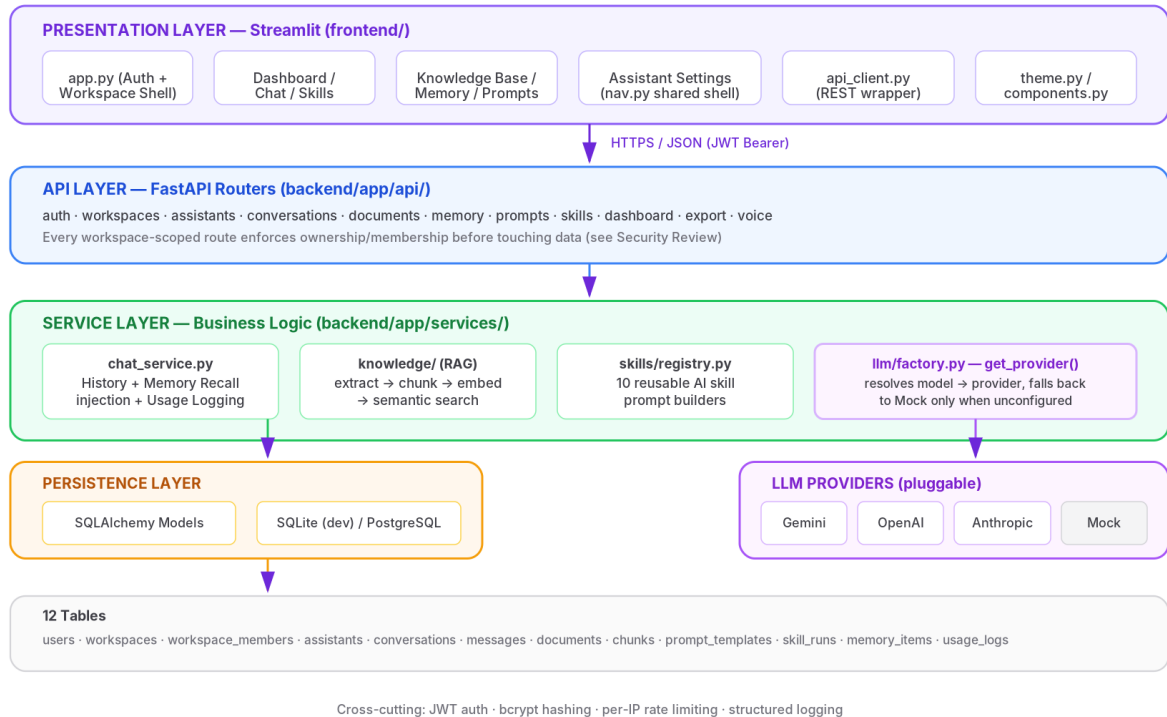


Figure 1. AI Workspace Platform Architecture

Architecture Explanation

Frontend / User Interface

The frontend provides the user-facing interface of the AI Workspace Platform. Users interact with the workspace, conversations, documents, prompts, memory, and AI skills through this layer.

Backend / API Layer

The backend acts as the central application layer. It receives requests from the frontend, processes application logic, communicates with the database, and coordinates AI-related operations.

Authentication and Authorization

Authentication and authorization controls are applied to protect access to the platform and its workspace resources. The backend verifies user access before allowing protected operations.

AI / LLM Provider Layer

The platform can communicate with an external AI provider such as Gemini for generating responses. The architecture also supports a mock provider for local or fallback operation when an external API key is unavailable.

Knowledge Base / Document Processing

Uploaded documents are processed and converted into chunks and embeddings. These representations support semantic search and document-grounded question answering.

Memory and Workspace Data

Workspace information, conversation data, prompts, memory items, and related application data are persisted in the platform's database.

Overall Data Flow

A typical request starts from the frontend and moves through the backend/API layer. The backend applies the required access checks and then communicates with the appropriate data or AI services. The resulting response is returned to the frontend for display to the user.